

ACR — Arc Compute Rate Submission Brief

Nine sections in the form's own order. Each opens with the literal text for that field, then the depth behind it. Every figure here is permanently on-chain or CI-gated — nothing on this page can drift between writing and judging.

10 Aug 2026 · arc-compute-rate.vercel.app · github.com/kaustubh76/ACR---Arc-Compute-Rate

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Project title

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ACR — Arc Compute Rate

Tagline, where the form allows one: *"Machine commerce just got its SOFR — and it prints its own attack cost."*

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Description

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ACR is "SOFR for machine commerce": a manipulation-resistant reference-rate family (inference, GPU, data) recovered from the noisy, adversarial payment exhaust on Circle's Arc L1 and published hourly on-chain — each print carrying its confidence interval and the USDC an attacker must burn to move it one basis point. The rate is sold to machines over x402 nanopayments, an on-chain futures venue cash-settles against it, and an autonomous agent on a Circle agent wallet buys the print and hedges with it, no human in the loop.

The loop, which is the whole product: **measure** (four-pillar estimator → EIP-712 print, signed by a Circle custody wallet) → **sell** (\$0.0001 per query over Circle Gateway x402) → **settle** (ACRFutures fills and cash-settles at that same print, refusing one older than two hours) → **hedge** (the agent pays for the number, reads its own book, trades the gap to its mandate). A benchmark is a number something settles against — and the settling contract breaks when the feed breaks, which is what separates this from a dashboard.

Track

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Agentic Economy

Fit, in one sentence: the autonomous hedger answers the track’s standing question — *does it actually work autonomously?* — with two separately checkable legs: its x402 payments are Gateway batch references on a public receipts tape, and its position is an Arc transaction (tx 0x22772154...), neither of which asks anyone to take the agent’s word.

Circle account email

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kaushtubhagrawal45@gmail.com

The developer-console account whose Developer-Controlled wallets sign every oracle print, own the futures venue, and stand behind the maker — so the API usage judges see on this account *is* the product running.

Products used

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Arc L1 (4 contracts, native USDC gas) · Circle Gateway x402 nanopayments (seller gate + buyers) · Developer-Controlled Wallets (press / maker / taker / treasury) · User-Controlled Wallets + Gas Station (Public Desk, PIN) · Agent Wallets + Circle CLI (autonomous hedger) · Unified Balance Kit (agent-run Gateway deposits) · Circle webhooks (signed inbound) · Agent Marketplace (Discovery-shaped catalog + public receipts tape; listing submitted) · Circle Skills (four consumed, one published back: acr-hedge)

PRODUCT	THE JOB IT DOES IN ACR	EVIDENCE
Arc L1	The chain itself: ACROracle, ACRFutures, AttestationRegistry, FeedAccessAttestor; USDC is the gas token, which makes the attack cost arithmetic instead of a distribution	chain 5042002 · testnet.arcsan.app
Gateway x402	Both sides of the payroll: the seller gate verifies and settles EIP-3009 payments; agents pay \$0.0001/query	36 settled queries · batch-UUID receipts, public tape

Developer-Controlled Wallets	The cast's payroll: the press signs hourly prints, the maker owns and collateralises the venue, the taker heartbeats it, the treasury funds them	signer 0x8366968f... · maker 0x9D44A7Dd...
User-Controlled + Gas Station	The Public Desk: a stranger opens an SCA in the browser with a PIN, trades the venue for real, withdraws — sponsorship proven from the ERC-4337 paymaster event	4 non-operator SCAs have traded; 2 cleared in settled rounds
Agent Wallets + Circle CLI	The hedger: SCA trades, backing EOA pays for prints (EIP-3009 needs ecrecover); Circle's own CLI paid the gate end-to-end	agent 0x1Dc707E3... · fill tx 0x22772154...
Unified Balance Kit	<code>make gateway-deposit</code> : the Gateway top-up an agent runs for itself instead of a human running the CLI	real 0.5 USDC deposit, tx 0xd6bab6ee...
Webhooks	Signed inbound receiver for Circle notifications	/webhooks/circle, signature-checked
Agent Marketplace	Discovery-shaped catalog of 13 priced resources + the public settlement tape the terminal's storefront buys from	/marketplace/catalog · listing submitted 04 Aug
Circle Skills	Consumed Circle's four; published one back — <code>skills/acr-hedge</code> teaches any agent the discover → pay → read venue → hedge loop	skills/acr-hedge/SKILL.md

Not used, said plainly: Smart Contract Platform (contracts deployed via Foundry; `deploy_circle.py` exists, is tested, and has never deployed — a tested code path is not a shipped one) and CCTP (single-chain by construction: the benchmark's dependents live where it prints, and a bridge would blur the one thing that makes the attack cost a number).

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Working MVP

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Live terminal: <https://arc-compute-rate.vercel.app>
Seller API (x402-gated): <https://acr-api-1fto.onrender.com>
Venue on Arc:
<https://testnet.arcscan.app/address/0x29d97c629a8278f7ec4218ab0bd8baa9182642fe>
Repo: <https://github.com/kaustubh76/ACR---Arc-Compute-Rate>
Reproduce: `make setup && make ci && make demo`

SLOT

Demo video (3:00): <add the link after recording — script at docs/DEMO-SCRIPT.md>

Ten-second checks a judge can run:

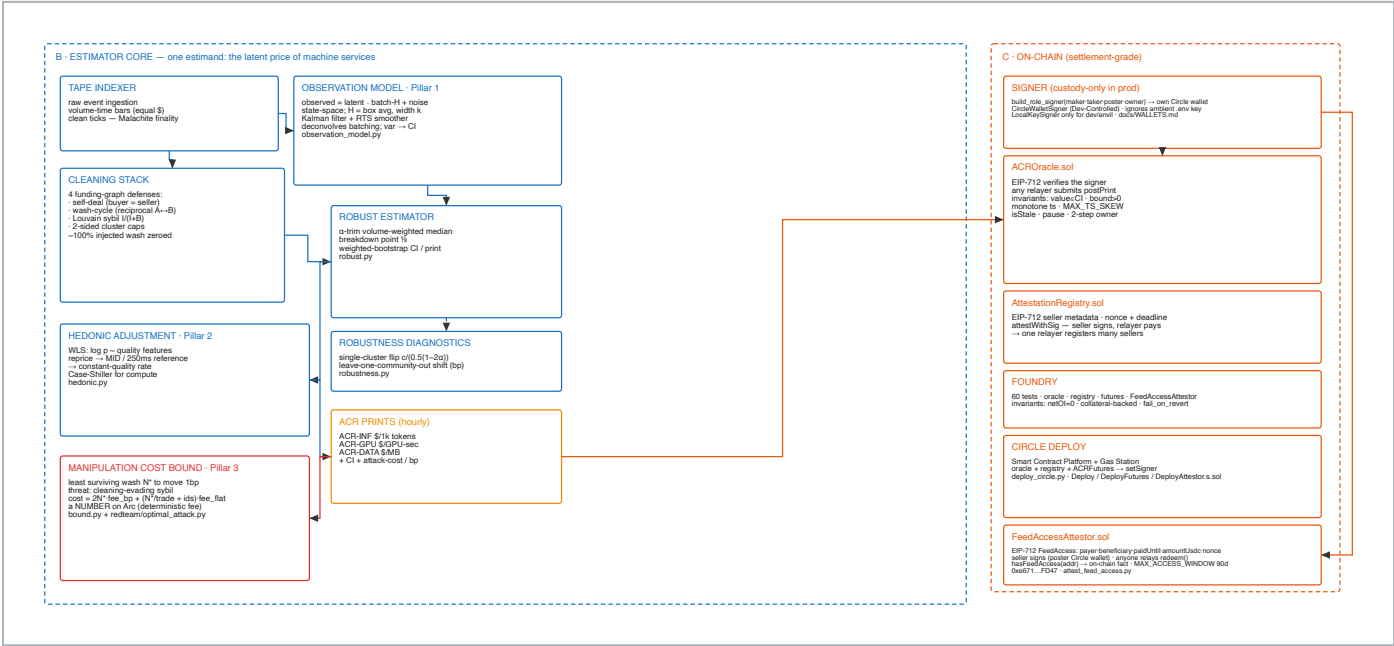
- **The site is live against Arc, not a recording:** /sellers → press *read it from the chain* twice — the block number lands, then lands *higher*.
- **Real settlement on camera:** /exchange → open a listing → *buy this one* · \$0.0001 — a real Gateway batch reference lands and the public tape gains a row.
- **The autonomous agent, both legs:** GET /hedger — its x402 receipts (off-chain batch refs) beside its on-chain position and fill.

Standing on-chain evidence (permanent, linkable):

FACT	VALUE
Three settled rounds	#2 @ 0.49533 (tx 0xf094befc...) · #1 @ 0.49533 (tx 0x5351bd0c...) · #0 @ 0.49270 — cash, final, collateral released; two cleared wallets are outside the operator set
The hedger's fill	buy 0.24 @ 0.49773 · tx 0x22772154ff8deb5ef43001af8c98fecdc36ae2dc70cfe437965c90f68d0041834 · position 2.47 → 2.71
Resistance (\$8,000 wash attack)	naive VWAP +110.8% vs ACR +0.20% → 562× ; 12-h eval: 5703.9 bp vs 124.1 bp = 46× ; CI-gated (make eval-gate)
Suites, all green	367 py · 60 forge · 95 terminal · 10 agent · 423/423 glossary · GitHub CI 4/4

Honesty, stated before it is asked: the price tape is a *labelled* simulator (tape_source: sim — ~18.5k real Arc settlements collapse to one price, so no index is honestly publishable from them yet), and every x402 payer so far is ours. The plumbing is proven; the demand is not. The site says this too.

Architecture diagram



Tape → clean (wash · sybil) → deconvolve (Kalman/RTS) → α-trim median + hedonic → print + CI + attack-cost → EIP-712 oracle → futures settle → agents buy the rate back via x402.

The full canvas with every box mapped to its module lives in docs/ARCHITECTURE-DIAGRAM.md — called implementation-accurate because CI holds it to that: the diagram’s suite counts are measured by make verify-claims and its 423 terms are all defined in the glossary (make glossary-check).

Documentation

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Judge’s one-pager: docs/SUBMISSION.md — every claim with its evidence and dated gates. Methodology: docs/methodology.md. Architecture: docs/ARCHITECTURE-DIAGRAM.md. API: docs/acr-openapi.md. How to run: IMPLEMENTATION.md (make setup && make ci && make demo).

DOCUMENT	ITS ONE JOB
README.md	The front door: what ACR is, demo-day metrics, how to use it
docs/SUBMISSION.md	The one-page status for judges — every claim beside its evidence, every gate dated
docs/methodology.md	The estimator spec — four pillars, published before liquidity, the way SOFR was
docs/ARCHITECTURE-DIAGRAM.md	The implementation-accurate blueprint behind the diagram above
docs/acr-openapi.md · .pdf	The API, endpoint by endpoint — mirrored live at /developers, where every free endpoint runs from the page

docs/WALLETS.md	Which Circle wallet product does which job, and the constraint that forced each choice
docs/agent-runbook.md	The Circle CLI path: create, fund, deposit, pay — proven end-to-end
docs/GLOSSARY.md	423 terms, CI-gated against the diagram — no undefined jargon anywhere
docs/TESTNET_RUNBOOK.md · DEPLOY.md	The ordered operator sequences: go-live on Arc, cloud deploy
docs/PITCH.md · pitch deck	The 8-slide deck (docs/pitch/, make pitch → PDF) and the spoken argument with objection handlers
docs/DEMO-SCRIPT.md · teleprompter	The 3-minute video, beat by beat, re-verified against production 10 Aug
skills/acr-hedge/SKILL.md	The published Circle Skill: teaches any agent the loop, with the two gotchas that cost us most

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Product feedback

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Everything below was measured while building on the full stack (Gateway x402, three wallet models, CLI, UBK, webhooks, Skills) on Arc testnet. Sharpest finding: `circle wallet execute` cannot BUILD a transaction carrying a negative int256 (eth_call succeeds; a real revert errors differently), so an agent on that path can open a position but never close one — one-directional autonomy. Full list with repro detail in the repo's CIRCLE-SESSION-QUESTIONS.md.

What worked, and impressed:

- Gateway nanopayment batching end-to-end** — 402 → EIP-3009 → batch settlement worked exactly as documented at \$0.0001; our 402 parses byte-identically in Circle's own GatewayClient (12/12 interop, re-run against the deployed gate).
- Gas Station** — sponsorship is cleanly observable from the ERC-4337 UserOperationEvent paymaster, which let us prove sponsorship instead of asserting it.
- Hosted PIN custody** — a stranger provisions an SCA and trades in a browser with the key never touching our server; the trust story explains itself on camera.
- Skills** — the plugin model was easy to consume and easy to publish back; acr-hedge took an afternoon.

Measured integration findings (each cost real debugging time; repro detail in CIRCLE-SESSION-QUESTIONS.md):

#	FINDING	WHY IT BITES
1	circle wallet execute cannot build a tx with a negative int256 — verified with controls (eth_call succeeds; an actual revert produces a different error)	An agent can open a position and never unwind it: one-directional autonomy

2	<code>services pay</code> buries the Gateway batch UUID in base64 inside <code>data.payment.receipt</code>	The one reference that ties a payment to its settlement is invisible to a script that reads the obvious fields
3	Circle APIs return lowercase addresses ; web3.py rejects non-checksummed input	The failure presents as throttling, not as a format error — hours lost in the wrong direction
4	The w3s SDK challenge callback never fires on an already-COMPLETED challenge	Resuming a PIN ceremony needs a status poll; the documented callback path silently hangs
5	The hosted PIN flow gates on literally typing "I agree" — undocumented	Stalls E2E and demo automation; nothing in the flow says why it is waiting
6	Sponsorship appears as a paymaster event topic, not in <code>networkFee</code>	Fee accounting that reads the obvious field mis-reports sponsored transactions
7	Arc <code>eth_getLogs</code> : ~15,000-block hard cap (413), string <code>toBlock</code> rejected, plus 429 throttling	Backing off the <i>range</i> on a 429 collapses read reach — a throttle must never be treated as a range error (measured: a 7.9 h walk collapsed to 1,264 blocks, ≈11 min)
8	Unified Balance Kit <code>getBalances</code> needs <code>chains:</code> <code>"Arc_Testnet"</code> spelled exactly	Without it, Arc silently reads 0 — indistinguishable from an unfunded wallet

The asks, in priority order:

- **Arc in the Discovery API.** Circle's marketplace serves 958 listings — every one mainnet, no Arc network at all — so an Arc-native seller meeting every listing prerequisite cannot be discovered by the very agents the track is about. Our listing is submitted and waiting on exactly this.
- **Platform-enforced spend policies for agent-held wallets** (per-tx caps, allowlisted contracts, rate ceilings) — today the guardrails around an autonomous key are all client-side, which means they are promises, not properties.
- **Event push over log polling.** The getLogs limits above make an agent's world-view a paged crawl; a webhook/stream for contract events on Arc would remove our largest source of operational fragility.
- **Cross-chain funding of an Arc seller through Gateway's unified balance** — the one rail this project doesn't use yet; it would let any judge fund a buyer from whatever testnet USDC they already hold.